

System Documentation

Electronic Equipment

BlueVision | New Generation

Interface to Non-MTU Monitoring System

J1939 interface

Functional Description

Installation and Commissioning Instructions

Application: Marine

E532819/00E



Power. Passion. Partnership.

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1 Safety

1.1 Important requirements for all products

Nameplate

The product is identified by a nameplate, model designation or serial number which must match the information on the title page of this manual.

Nameplate, model designation or serial number can be found on the product.

General information

This product may pose a risk of injury or damage in the following cases:

- Improper use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications
- Noncompliance with the safety instructions and warning notices

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the permissible operating parameters in accordance with the (→ Technical data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/MTU contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (including engine management/parameters)
- In compliance with all safety regulations and in accordance with all warning notices in this manual
- With maintenance work performed in accordance with the (→ Maintenance Schedule) throughout the useful life of the product
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair
- By contracting only workshops authorized by the manufacturer to carry out repair and overhaul

Any other use shall be considered non-intended. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

Changes or modifications

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

1.2 Personnel and organizational requirements

Organizational measures of the operator

This manual must be issued to all personnel involved in operation, maintenance, repair or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair and transport personnel at all times.

Use this manual as a basis for instructing personnel on product operation and repair, whereby the safety-relevant instructions, in particular, must be read and understood.

This is particularly important in the case of personnel who only occasionally perform work on or around the product. This personnel must be instructed repeatedly.

Personnel requirements

All work on the product shall be carried out by trained and qualified personnel only.

- Training at the Training Center of the manufacturer
- Qualified personnel specialized in mechanical and plant engineering

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair and transport.

Working clothes and personal protective equipment

Wear proper protective clothing for all work.

When working, always wear the necessary personal protective equipment (e.g. ear protectors, protective gloves, goggles, breathing protection). Observe the information on personal protective equipment in the respective activity description.

1.3 Safety requirements for startup and operation

Safety requirements for startup

The product must be installed and accepted in accordance with manufacturers' specifications prior to initial startup.

All the requisite regulatory approvals must have been granted and all startup requirements fulfilled prior to initial startup.

Whenever the product is started, always ensure that:

- All maintenance and repair work has been completed.
- All loose parts have been removed from rotating machine components.
- Wearers of cardiac pacemakers or other active medical implants are well clear.
- The operating room is adequately ventilated.
- Exhaust pipework is leak-tight and routed to atmosphere.
- Battery terminals, generator terminals and cables are guarded to preclude accidental contact.
- All personnel is clear of the danger zone represented by moving parts.

Immediately after putting the product into operation, make sure that all control and display instruments as well as the monitoring, signaling and alarm systems are working properly.

Safety requirements for operation

The operator must be familiar with the controls and displays.

The operator must be aware of the consequences of any operations he/she performs.

During operation, the display instruments and monitoring units must be constantly observed in regard of present operating status, limit value violation and warning or alarm messages.

Malfunctions and emergency stop

Emergency procedures, in particular emergency stop, must be practiced on a regular basis.

Take the following steps if a system malfunction is detected or signaled by the system:

- Inform the duty supervisor(s).
- Evaluate the message.
- Respond to the emergency appropriately, e.g. execute an emergency stop.

Operation

Do not stay in the operating room when the product is running unless absolutely necessary and then as briefly as possible.

Keep a safe distance away from the product whenever possible. Do not touch the product unless expressly instructed to do so.

Do not inhale exhaust gases emitted by the product.

Ensure that the following requirements have been fulfilled before starting the product:

- Ear protectors are worn.
- Mop up any leaked or spilled fluids and lubricants immediately or soak up with a suitable binding agent.

Operation of electrical equipment

Some components are live (high voltage) when electrical equipment is in operation.

Observe the safety instructions for these appliances.

1.4 Standards for safety notices in the text

DANGER 	In the event of immediate danger. Consequences: Death, serious or permanent injury! • Remedial action.
WARNING 	In the event of a situation involving potential danger. Consequences: Death, serious or permanent injury! • Remedial action.
CAUTION 	In the event of a situation involving potential danger. Consequences: Minor or moderate injuries! • Remedial action.
NOTICE 	In the event of a situation involving potentially adverse effects on the product. Consequences: Material damage! • Remedial action. • Additional product information.

Safety notices

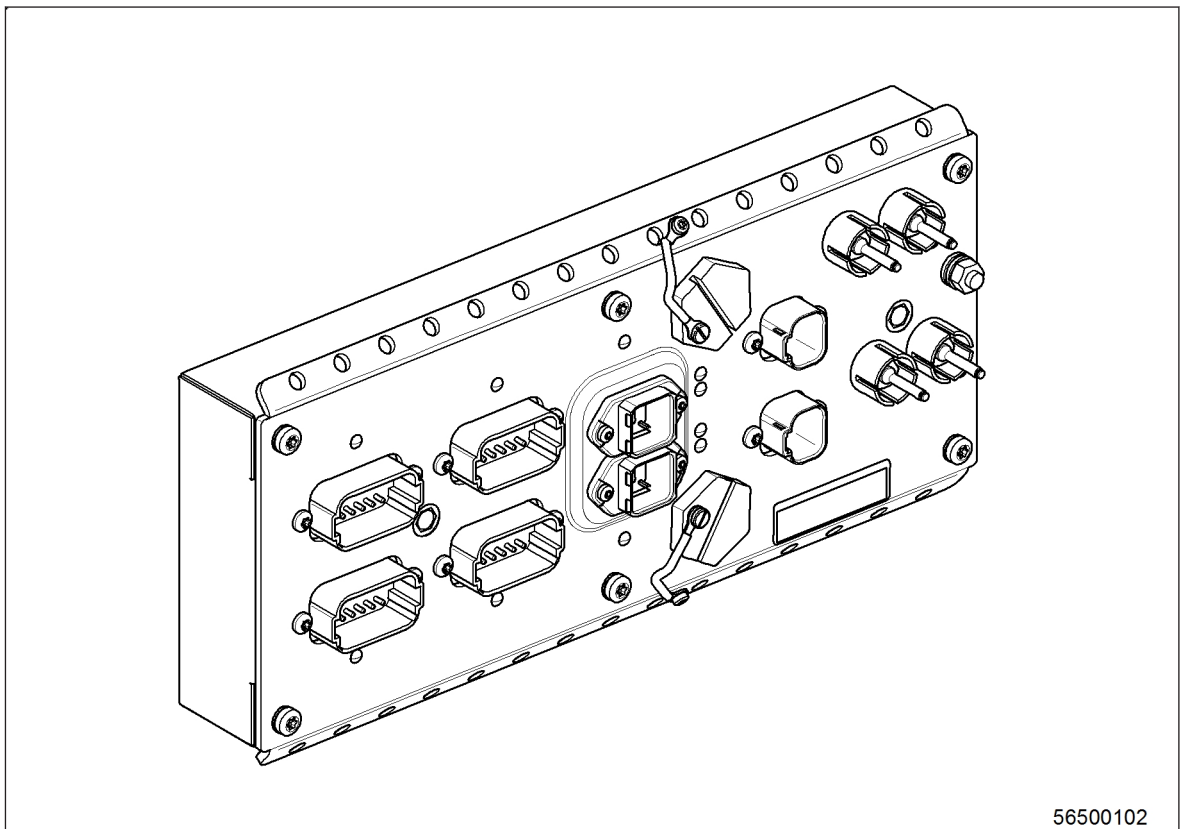
1. This manual with all safety instructions and safety notices must be issued to all personnel involved in operation, maintenance, repair or transportation.
2. The higher level warning notice is used if several hazards apply at the same time. Warnings related to personal injury shall be considered to include a warning of potential damage.

2 Functional Description

2.1 Systembus Coupling Unit SCU – Function

Systembus Coupling Unit SCU is a CAN-over-Ethernet (CoE) interface serving to link up the MTU system bus and the CAN bus of the remote control system. The unit features two independent processing units each of which providing two CAN interfaces (1/2). Input and output data can be transmitted via the CAN interfaces. The unit supports the SAE J1939 CAN protocol only.

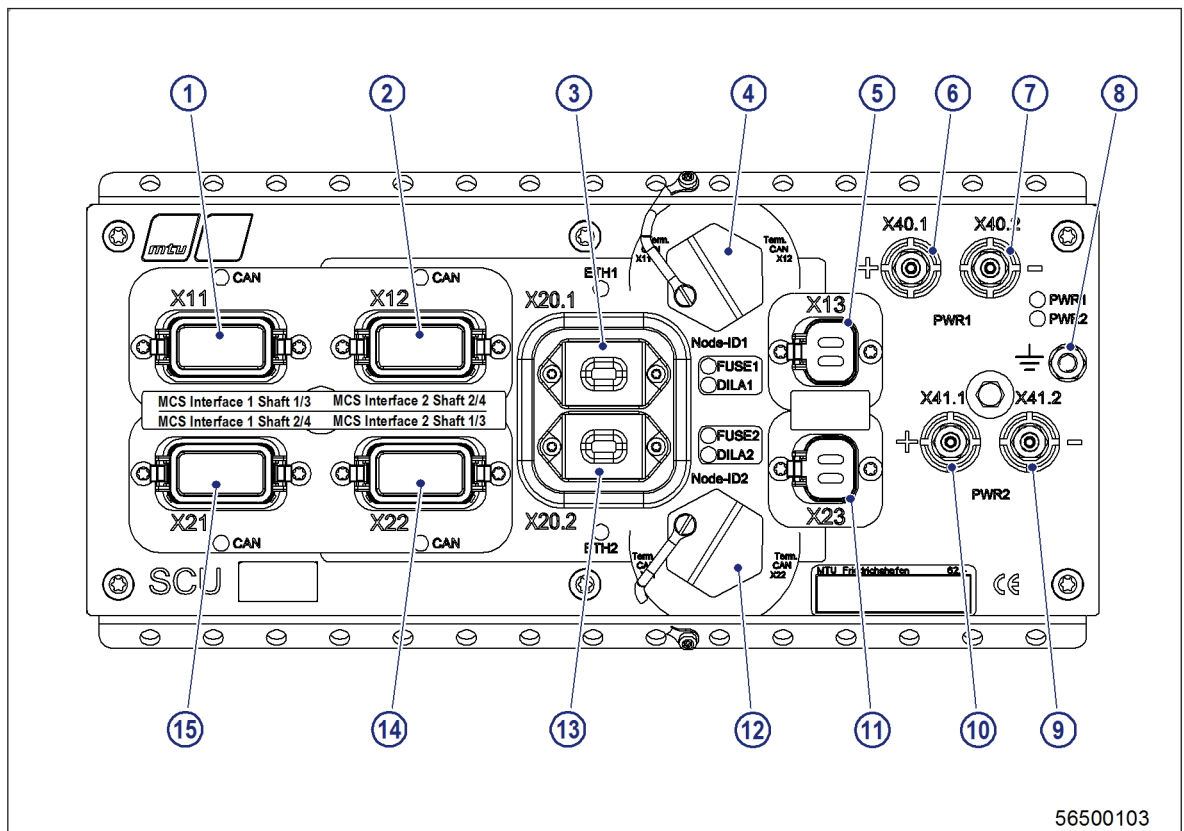
The two processing units exchange transmitted and received CAN packages with the other stations on the system bus. The two processing units can operate on a redundant power supply.



The SCU has the following tasks:

- Communication interface between the MTU system and the non-MTU monitoring system
- Providing two CAN data interfaces for engines 1 to 4:
 - J1939 data interface 1 and 2 for engine 1 (or engine 3)
 - J1939 data interface 1 and 2 for engine 2 (or engine 4)

2.2 Systembus Coupling Unit SCU - Design



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|------------------------------|-------------------------------|-------------------------------|
| 1 X11 CAN 1 (interface 1) | 6 X40.1 +24 V power supply 1 | 11 X23 not used |
| 2 X12 CAN 2 (interface 2) | 7 X40.2 GND power supply 1 | 12 Service port 2 |
| 3 X20.1 Systembus Ethernet 1 | 8 Grounding stud | 13 X20.2 Systembus Ethernet 2 |
| 4 Service port 1 | 9 X41.2 GND power supply 2 | 14 X22 CAN 4 (interface 2) |
| 5 X13 not used | 10 X41.1 +24 V power supply 2 | 15 X21 CAN 3 (interface 1) |

Processing module 1 forms a complete unit with the following SCU components:

- CAN interfaces X11 and X12
- Service port 1
- Associated diagnostic LEDs

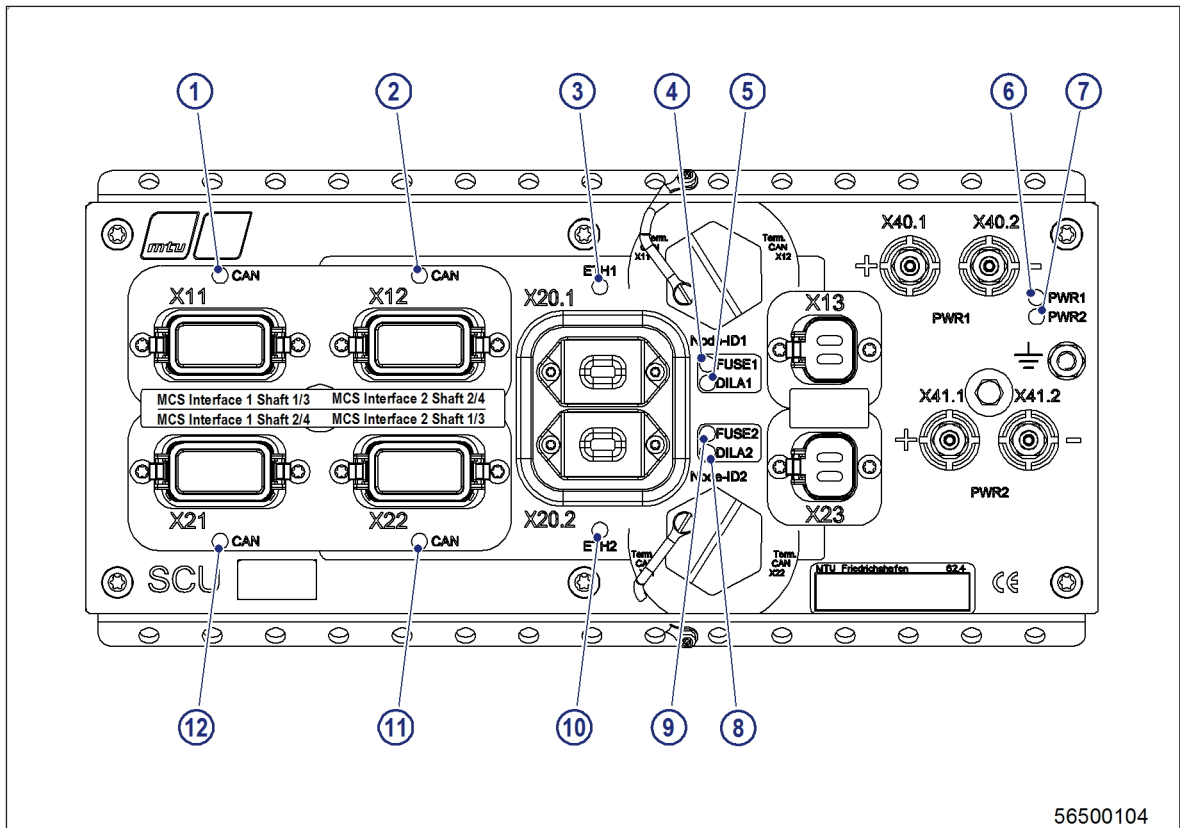
Processing module 2 also represents a complete unit with the following SCU components:

- CAN interfaces X21 and X22
- Service port 2
- Associated diagnostic LEDs

Both processing modules are independently connected to the system bus via a separate 3-port Ethernet switch.

The hexadecimal rotary coding switches and slide switches beneath service ports 1 and 2 can be used to set the system bus node number and connect the CAN bus termination resistors.

Signal lamps on SCU



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|-----------------------------------|----------------------------------|------------------------------------|
| 1 LED CAN 1 | 5 DILA 1 | 9 LED FUSE 2 – Not used |
| 2 LED CAN 2 | 6 LED PWR 1 +24 V power supply 1 | 10 LED ETH2 – Systembus Ethernet 2 |
| 3 LED ETH1 – Systembus Ethernet 1 | 7 LED PWR 2 +24 V power supply 2 | 11 LED CAN 4 |
| 4 LED FUSE 1 – Not used | 8 DILA 2 | 12 LED CAN 3 |

Systembus Coupling Unit SCU is equipped with several LEDs to indicate its operating state.

Diagnostic lamps DILA1 and DILA2 indicate the operating state of the two SCU processors. CAN bus data traffic or faults are indicated by the CAN LEDs.

System bus data traffic or faults are indicated by the system bus LEDs (LED-ETH1/2).

LED status indicatorsSCU

Item	Labeling	LED color	LED state	Meaning
1	CAN 1	Yellow	Flashing	CAN communication in progress.
		Red	Flashing	CAN fault.
		—	Dark	No CAN communication.
2	CAN 2	Yellow	Flashing	CAN communication in progress.
		Red	Flashing	CAN fault.
		—	Dark	No CAN communication.
3	ETH 1	Yellow	Flashing	LAN communication in progress.
		—	Dark	No LAN communication.
4	FUSE 1	—	—	Not used.

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